

important hardware components in a superconducting quantum computer—the Josephson junction. In the previous article, we decided to focus most of our hardware discussions on superconducting qubits, since they currently represent the most mature and commercially successful quantum computing technology. However, as we also noted, there is no guarantee that superconducting qubits will ultimately emerge as the winning architecture among the many competing quantum technologies. Nevertheless, understanding the Josephson junction provides valuable insight into how today's superconducting quantum processors, such as those developed by IBM and Google, actually work. It is no exaggeration to say that the Josephson junction is the heart of these quantum processors.

A Josephson junction consists of two superconducting materials separated by an extremely thin insulating layer, typically only a few nanometres thick. Although the insulating layer should, in principle, prevent electric current from flowing, quantum mechanics allows pairs of electrons, known as Cooper pairs, to tunnel through the barrier. This remarkable phenomenon, known as the Josephson effect, was predicted by the British physicist Brian D. Josephson in 1962 and later earned him the Nobel Prize in Physics. By the third article in this series, it has almost become a cliché to point out that the history of quantum computing is filled with Nobel laureates!

It is important to understand that programmers never interact with the Josephson junction directly. When

we write a Qiskit statement such as `qc.x(0)` or `qc.h(0)`, we are specifying a logical quantum operation rather than controlling the hardware itself. The quantum control system automatically translates these instructions into precisely calibrated microwave pulses, which manipulate the superconducting qubit through the Josephson junction. Thus, what appears as a single line of Qiskit code corresponds, at the hardware level, to an intricate sequence of physical processes occurring inside a quantum processor operating at temperatures of only a few millikelvin.

Although Qiskit focuses primarily on programming quantum circuits, there are specialised open source tools for modelling and designing superconducting quantum hardware. QuTiP (Quantum Toolbox in Python) is widely used to simulate the dynamics of quantum systems, including superconducting qubits and Josephson-junction-based circuits. Quantum Metal (formerly Qiskit Metal) is another open source framework that enables researchers to design and analyse superconducting quantum devices before they are fabricated. Although we will explore these tools in much greater detail later in this series, it is worthwhile to get a brief glimpse of QuTiP.

Consider the following sample program, which constructs the Hamiltonian of a simple two-level quantum system—the basic mathematical model used to describe a superconducting qubit.

```
from qutip import sigmax, sigmaz

omega = 5.0
delta = 1.0
H = 0.5 * omega * sigmaz() + 0.5 * delta * sigmax()
print(H)
```

The program above is not a complete physical simulation of a Josephson-junction circuit, but rather the simplest Hamiltonian that captures the behaviour of a superconducting qubit. More realistic Josephson-junction models involve additional physical parameters, non-linear circuit elements, and time-dependent microwave control pulses. We shall postpone these fascinating topics until a much later article in this series, where we will explore quantum hardware in greater depth using tools such as QuTiP and Quantum Metal.

So far our discussion has been confined to single-qubit systems. A lone qubit, fascinating though it is, cannot demonstrate the true power of quantum computing. That power begins to emerge only when multiple qubits interact with one another. In the next article, we will leave the world of isolated qubits behind and explore the CNOT gate, one of the most important quantum gates, which serves as the gateway to one of the strangest and most remarkable phenomena in modern physics—quantum entanglement, famously described by Albert Einstein as “spooky action at a distance.” **END** 

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